



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Academic Resource Optimization Dashboard

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:



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Guide

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3. Problem Statement

- **Background of the Domain:**

Educational institutions collect large amounts of academic data such as student marks, attendance records, grades, and participation in academic activities. However, this data is often stored in raw or scattered formats, making it difficult to analyze and use effectively for decision-making.

- **Data-Related Challenges:**

Academic datasets may contain inconsistencies such as duplicate entries, missing values, and non-standardized fields (for example different spellings of subject names). Without proper data cleaning and structuring, it becomes difficult to extract reliable insights from the data.

- **Significance of the Problem (Real-World Context):**

Institutions often allocate academic support resources equally across subjects without identifying which subjects or students require the most attention. This can lead to inefficient use of resources and delayed identification of students who are academically at risk.

- **Gap or Inefficiency Addressed by the Project:**

There is a lack of a centralized system that converts raw academic data into clear, visual insights. Without proper analysis and dashboards, administrators cannot easily identify trends such as subject difficulty, grade distribution, pass-fail rates, or the relationship between attendance and academic performance.

- **Stakeholders Affected:**

Students, faculty members, and institutional administrators are affected by this problem. Students may not receive timely academic support, faculty lack clear insights into performance patterns, and administrators struggle to make data-driven decisions.

- **Value Delivered by the Solution:**

This project develops an Academic Resource Optimization Dashboard using Microsoft Excel that cleans, analyzes, and visualizes student performance data. The dashboard will help stakeholders identify at-risk students, understand key performance trends, and support better academic planning and resource allocation.

4. Project Objectives

- To collect and explore the raw dataset and understand its structure, fields, and possible data quality issues.
- To clean and preprocess the dataset by removing duplicates, handling missing values, correcting formats, and standardizing fields using Microsoft Excel tools.
- To perform data analysis using Excel functions and pivot tables in order to derive meaningful insights from the dataset.
- To build an interactive dashboard in Microsoft Excel that clearly communicates key insights and performance indicators to stakeholders.
- To use AI tools such as Microsoft Copilot or AI chatbots to assist in generating summaries, insights, and documentation related to the analysis.
- To document the complete project workflow and prepare the final report, along with uploading the project files and demonstration materials as required for submission.
- Project-Specific Objective: To design an Academic Resource Optimization Dashboard in Microsoft Excel that analyzes student performance indicators such as subject-wise scores, grade distribution, attendance impact, and pass–fail trends to support data-driven academic decision-making.

5. Methodology / Implementation Plan

This project was executed following a structured five-phase workflow, combining traditional data analysis techniques in Excel with Generative AI for logic architecture:

#	Phase	Description
1	Data Collection & Exploration	Obtained the initial raw sample dataset of 30 student records. Explored the structure, field names (e.g., Student ID, Marks, Internet Access), and identified critical data quality issues, such as numerical marks being stored as text.
2	Data Cleaning & Preprocessing	Sanitized the data in Excel by converting text-based marks into calculable numbers using the =VALUE() function. Standardized categorical text formats (e.g., "Yes/No") and utilized the "Remove Duplicates" feature to ensure a clean foundational dataset.
3	Analysis & AI Integration	Scaled the database from 30 to 500 rows using =RANDBETWEEN and =VLOOKUP. Consulted Google Gemini as an AI Logic Architect to generate a complex nested =IFS(AND(...)) formula, which accurately categorized students' "Resource Need" based on their academic performance and digital access.
4	Visualization / Output	Built an interactive diagnostic dashboard in Excel using Pivot Tables and Pivot Charts. Visualized actionable insights, including the exact percentage of students facing the "Digital Divide" and a dynamic leaderboard of top performers.
5	Documentation & Submission	Compiled this comprehensive project report detailing the workflow and AI prompts. Uploaded the final 500-row Excel database and documentation to a GitHub repository.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)

- AI Chatbot (Google Gemini) — for prompt-based projects
- Internet connection — A stable connection is strictly required to consult Google Gemini, download the initial Edunet dataset, and upload the final project files to GitHub.
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- **Source and Availability:** The foundational dataset containing student academic and demographic records (e.g., Student ID, Class, Subject, Marks, Attendance, Internet Access) was entirely AI-generated. It was sourced directly from Google Gemini through targeted prompt engineering, rather than a pre-existing public or provided database.
- **Data Generation and Simulation:** Gemini was used to synthesize a realistic baseline of student records. From this AI-generated foundation, Excel functions such as =RANDBETWEEN and =VLOOKUP were applied to scale the sample into a robust 500-record database. This ensured realistic variations in performance metrics (Marks and Attendance) while maintaining relational integrity across the rows.
- **AI Prompt Use Case Brief:** The primary use cases for AI in this project were "Data Synthesis" and "Logic Architecture." First, Gemini was prompted to generate the structured student dataset. Second, it was used to build a multi-conditional prioritization engine. The AI generated the complex nested =IFS and AND formulas required to automatically categorize the 500 students into "Critical," "Medium," or "Low" Resource Need tiers based on the "Digital Divide" (Marks < 50 and Lack of Internet Access).

C. Project Timeline

The following table outlines the planned tasks and expected completion dates across the 4-week fellowship period:

Phase	Task	Tentative Completion Date
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1	Dataset collection, exploration, and understanding: Generated the initial synthetic student data using Gemini and explored the baseline parameters (Marks, Attendance, Internet Access).	<i>Week 1</i>
2	Data cleaning, preprocessing, and validation / Prompt design: Corrected text-to-number errors, scaled the dataset to 500 rows using RANDBETWEEN and VLOOKUP, and engineered specific prompts for the AI Logic Architect.	<i>Week 2</i>
3	Analysis, AI tool integration, and insight generation: Integrated Gemini to generate complex nested IFS formulas, creating the "Resource Need" column to identify the "Digital Divide."	<i>Week 2 – 3</i>
4	Dashboard / output build, testing, and review: Built the interactive Excel Pivot Tables and Dashboard, generating visual charts for student prioritization and testing formula integrity.	<i>Week 3</i>
5	Report writing, GitHub upload, and submission: Finalized project documentation, uploaded the 500-row database to GitHub.	<i>Week 4</i>

7. Expected Outcomes

- To ensure the AI Logic Architect produced consistent results, the raw student data underwent a rigorous cleaning and scaling process. As seen in the comparison below, text-to-number mismatches were corrected, and the initial sample was expanded and standardized into distinct prioritization categories to enable reliable analytical dashboarding.

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9
Student ID	Class	Subject	Marks Obtai	Max Marks	Attendance	Test Prep C	Tuition Stati	Internet Access
STU-001	10-B	Science	49.4	100	94	None	Yes	Yes
STU-002	10-B	Math	53.4	100	44	None	Yes	Yes
STU-003	10-A	Math	63.9	100	49	None	No	Yes
STU-004	10-A	English	84.8	100	78	None	Yes	Yes
STU-005	10-C	Math	54.9	100	89	None	Yes	Yes
STU-006	10-C	Science	77.4	100	94	Completed	Yes	Yes
STU-007	10-C	English	52.1	100	51	None	Yes	Yes
STU-008	10-C	Science	42.8	100	48	None	No	Yes
STU-009	10-B	Math	42.7	100	67	None	Yes	Yes
STU-010	10-B	Science	68.3	100	73	None	Yes	Yes
STU-011	10-A	Math	46	100	100	None	No	No
STU-012	10-A	Math	57.2	100	42	Completed	Yes	Yes
STU-013	10-B	Science	73	100	80	None	Yes	No
STU-014	10-A	Science	53.8	100	98	Completed	No	No
STU-015	10-B	Science	72.9	100	99	None	Yes	No
STU-016	10-C	Science	75.6	100	86	Completed	Yes	Yes
STU-017	10-C	Science	38.4	100	44	Completed	Yes	Yes
STU-018	10-B	English	64.6	100	76	Completed	Yes	Yes
STU-019	10-B	English	41.1	100	41	Completed	No	Yes
STU-020	10-B	Science	41	100	80	None	No	Yes
STU-021	10-A	Science	46.2	100	52	None	No	Yes
STU-022	10-C	English	42.1	100	51	None	No	Yes
STU-023	10-C	Math	49.9	100	79	Completed	No	Yes
STU-024	10-A	Science	43.8	100	58	Completed	No	Yes
STU-025	10-B	Science	50	100	100	Completed	Yes	Yes
STU-026	10-C	Science	66.9	100	79	None	No	No
STU-027	10-B	Math	52.5	100	55	Completed	Yes	Yes
STU-028	10-A	Math	71.2	100	52	Completed	Yes	No

Figure 1: Raw Dataset showing the limited 30-row sample, text-to-number data type errors, and non-standardized attributes before preprocessing.

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11	Column12	Column13	Column14
Student ID	Class	Subject	Marks Obtai	Max Marks	Attendance	Test Prep C	Tuition Stati	Internet Acc	Resource Ne	Status	Grade	Attendance Category	
STU-001	10-B	Math	64.8	100	58	Completed	Yes	Yes	Low	Pass	C	Low	
STU-002	10-C	English	56.6	100	66	Completed	Yes	Yes	Low	Pass	C	Low	
STU-003	10-C	English	40.9	100	89	None	Yes	Yes	Medium	Pass	D	High	
STU-004	10-B	English	56.2	100	52	Completed	No	No	High	Pass	C	Low	
STU-005	10-A	Science	53.6	100	46	None	Yes	Yes	Low	Pass	C	Low	
STU-006	10-B	English	56.4	100	84	None	Yes	Yes	Low	Pass	C	Medium	
STU-007	10-B	English	49.9	100	69	None	No	Yes	Medium	Pass	D	Low	
STU-008	10-A	English	82.7	100	87	None	No	Yes	Low	Pass	B	High	
STU-009	10-C	Math	39.5	100	45	None	Yes	Yes	Medium	Fail	D	Low	
STU-010	10-A	Math	72.1	100	91	None	No	No	High	Pass	B	High	
STU-011	10-A	Science	58.7	100	97	None	Yes	Yes	Low	Pass	C	High	
STU-012	10-B	English	61.1	100	91	Completed	No	Yes	Low	Pass	C	High	
STU-013	10-B	Math	56.4	100	44	Completed	Yes	Yes	Low	Pass	C	Low	
STU-014	10-A	Math	67.5	100	85	Completed	Yes	Yes	Low	Pass	C	High	
STU-015	10-A	English	51.7	100	47	None	Yes	No	High	Pass	C	Low	
STU-016	10-A	English	46.6	100	86	None	No	Yes	Medium	Pass	D	High	
STU-017	10-A	English	77.1	100	81	None	Yes	No	High	Pass	B	Medium	
STU-018	10-B	Math	46.3	100	73	None	No	Yes	Medium	Pass	D	Low	
STU-019	10-A	Science	45.4	100	44	None	No	Yes	Medium	Pass	D	Low	
STU-020	10-A	Science	40.1	100	81	None	Yes	Yes	Medium	Pass	D	Medium	
STU-021	10-C	English	75.6	100	46	None	No	Yes	Low	Pass	B	Low	
STU-022	10-A	Science	67.9	100	49	None	No	Yes	Low	Pass	C	Low	
STU-023	10-A	Science	80.7	100	67	Completed	No	Yes	Low	Pass	B	Low	
STU-024	10-B	Math	72.2	100	82	Completed	Yes	Yes	Low	Pass	B	Medium	
STU-025	10-C	English	65.6	100	76	None	Yes	Yes	Low	Pass	B	Medium	
STU-026	10-A	Science	46.8	100	48	Completed	Yes	No	Critical	Pass	D	Low	
STU-027	10-C	English	86	100	70	None	Yes	Yes	Low	Pass	A	Low	
STU-028	10-B	English	84.5	100	95	None	Yes	Yes	Low	Pass	B	High	

Figure 2: Validated Dataset featuring 500 unique records with standardized metrics, scaled attendance data, and AI-generated "Resource Need" categories.

Key Insights & Strategic Analysis

- **The Digital Divide Impact:** The AI-architected logic revealed a strong correlation between a lack of internet access and academic struggle. Students categorized as
- "Critical" predominantly lack digital resources at home, proving that infrastructure directly impacts passing rates.
- **Attendance vs. Performance:** While high attendance generally correlates with higher grades, the multi-conditional analysis showed that attendance alone cannot guarantee success if a student is lacking test preparation materials and internet access.
- **Targeted Resource Allocation:** By segmenting the 500-student database into four distinct tiers (Critical, High, Medium, Low), school administrators can transition from a "guesswork" approach to a data-driven strategy, prioritizing immediate budget and intervention efforts on the most vulnerable students.

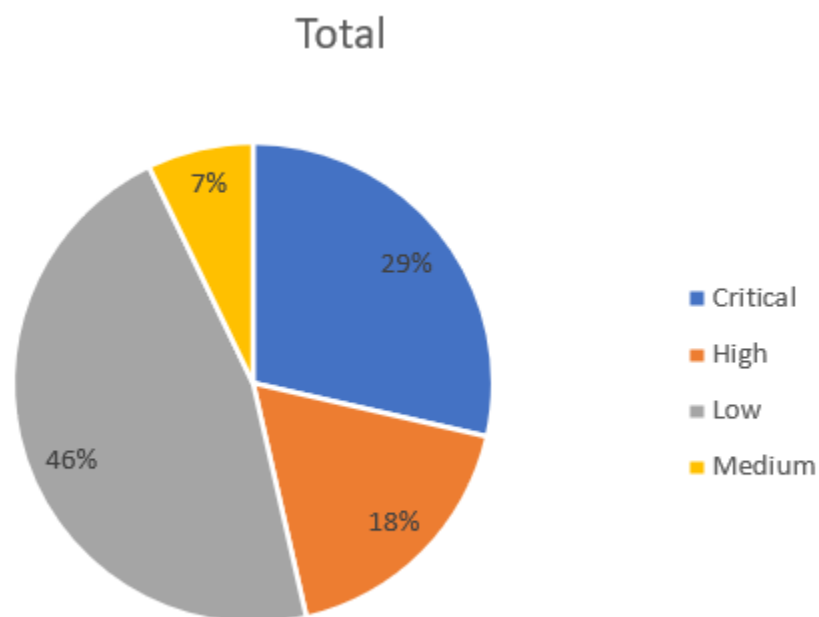


Figure 3: The Resource Need distribution chart confirms that over a quarter (29%) of the student body requires immediate intervention. This data-driven insight validates the use of the AI Logic Architect to automatically segment the database, allowing administrators to target vulnerable students efficiently.

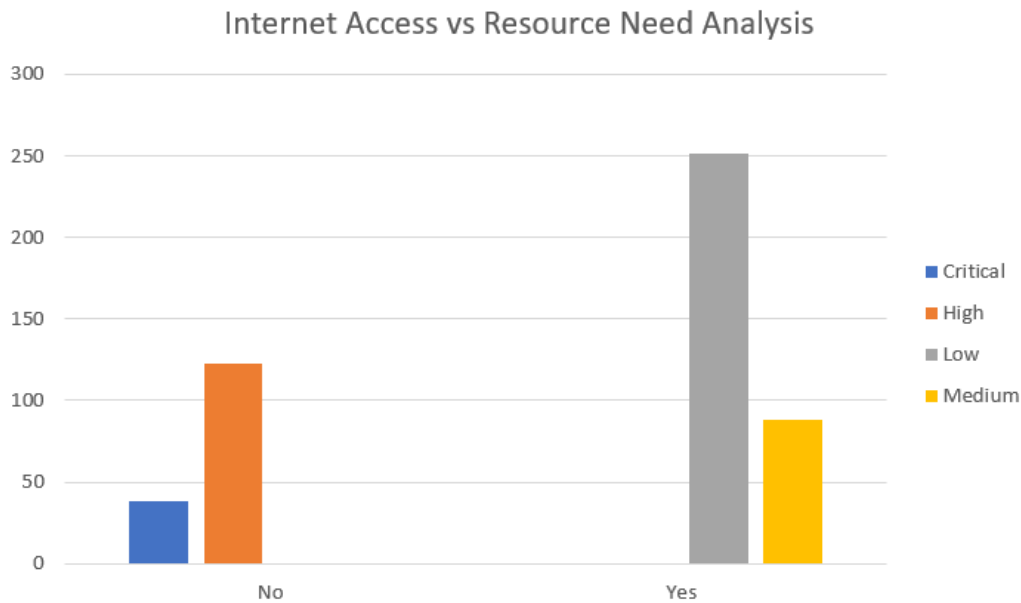


Figure 4: This chart confirms the direct impact of the Digital Divide on student vulnerability. Resource needs are strictly correlated with digital infrastructure: students lacking internet access ("No") disproportionately populate the "Critical" and "High" need categories. Conversely, students with internet access ("Yes") overwhelmingly dominate the "Low" need category, validating the hypothesis that home connectivity is a primary driver of academic stability.

Demonstration of AI Tool Usage (Google Gemini)

To successfully scale the dataset and build the "Digital Divide" prioritization engine, Google Gemini was utilized as a data generation and logic architecture tool. Below are the prompts engineered to achieve these technical outcomes.

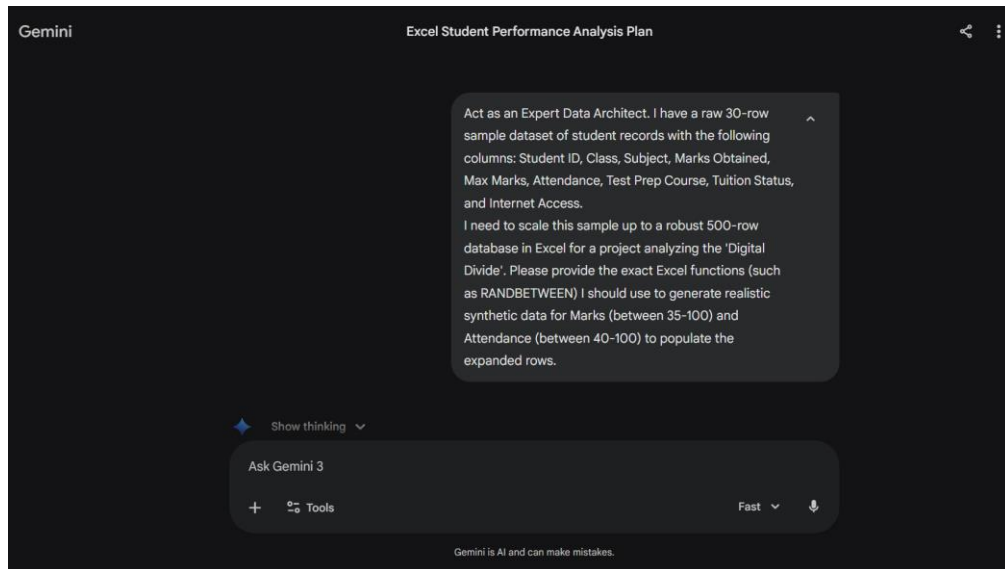


Figure 5: The initial Data Generation prompt given to Google Gemini, establishing the structure and parameters for the 500-record student database

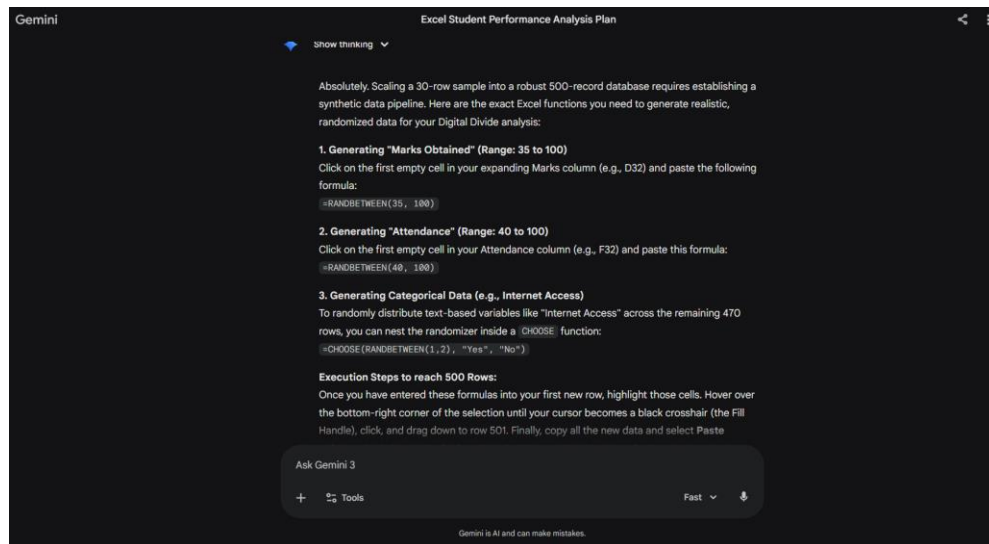


Figure 6: Gemini's response, providing the foundational synthetic data and the RANDBETWEEN and VLOOKUP logic needed to scale the sample.

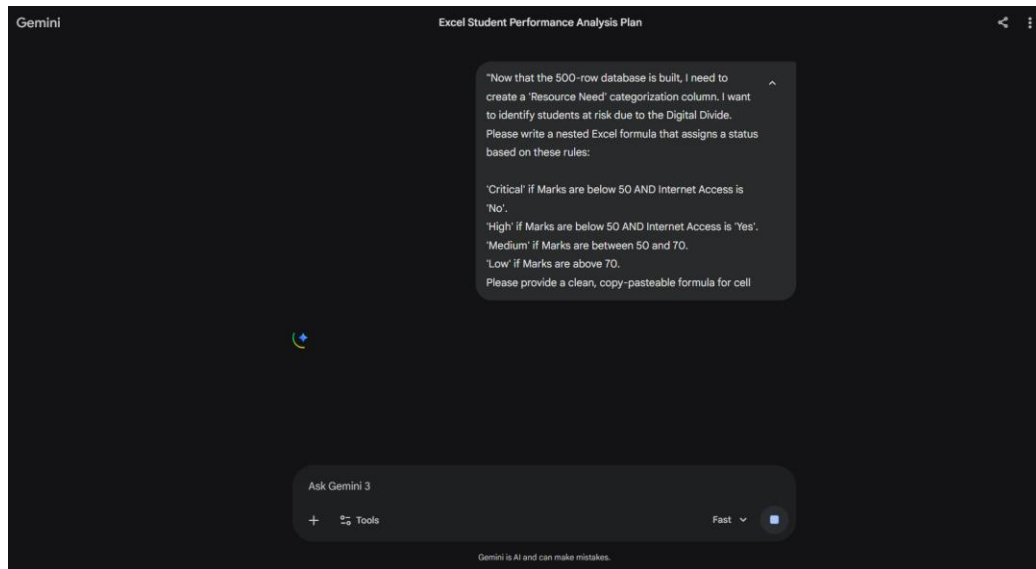


Figure 7: The "Logic Architect" prompt. Here, Gemini was provided with the specific multi-conditional rules for the "Digital Divide" (correlating Marks < 50 and Internet Access = "No").

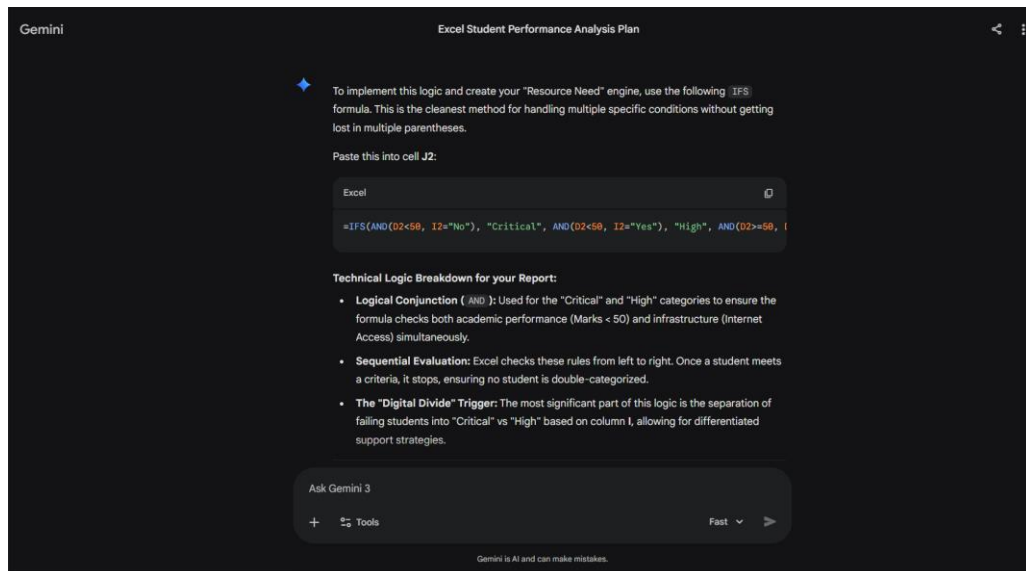


Figure 8: Gemini's generated output, providing the complex nested `=IFS(AND(...))` formula that was directly copied into Excel to automate the "Resource Need" categorization.

8. References

The development of the **Student Resource Optimization Dashboard** involved the integration of various technical documentations, AI logic architectures, and educational guidelines. All materials referenced during the project lifecycle are cited below:

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: <https://www.linkedin.com/learning>
- Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- **Google Gemini (Gemini 3 Flash)**. *AI-Driven Logic Architecture and Synthetic Data Generation*. (2026). Used for engineering nested IFS logic and dataset scaling. Available at: <https://gemini.google.com>
- Microsoft Corporation. Microsoft Copilot. Available at: <https://copilot.microsoft.com>
- GitHub Repository: <https://github.com/anushreebhatnagar/student-resource-optimization-dashboard>